17. Which of the following vitamins is water soluble?

a) Vitamin E b) Vitamin K c) Vitamin A d) Vitamin B

18. Complete hydrolysis of cellulose gives

a) L-Glucose b) D-Fructose c) D-Ribose d) D-Glucose

19. Which of the following statement is not correct?

a) Ovalbumin is a simple food reserve in egg-white

b) Blood proteins thrombin and fibrinogen are involved in blood clotting

c) Denaturation makes protein more active

d) Insulin maintains the sugar level of in the human body.

20. Glucose is an aldose. Which one of the following reactions is not expected with glucose?

a) It does not form oxime

b) It does not react with Grignard reagent

c) It does not form osazones

d) It does not reduce tollens reagent

21. If one strand of the DNA has the sequence 'ATGCTTGA', then the sequence of complementary strand would be

a) TACGAACT b) TCCGAACT c) TACGTACT d) TACGRAGT

22. Insulin, a hormone chemically is

a) Fat b) Steroid c) Protein d) Carbohydrates

23. α-D (+) Glucose and β-D (+) glucose are

a) Epimers b) Anomers

c) Enantiomers d) Conformational isomers

24. Which of the following are epimers

a) D(+)-Glucose and D(+)-Galactose (b) D(+)-Glucose and D(+)-Mannose

c) Neither (a) nor (b) (d) Both (a) and (b)

25. Which of the following amino acids are achiral?

a) Alanine b) Leucine c) Proline d) Glycine

Short Answer Questions

1. What type of linkages hold together monomers of DNA?
2. Give the differences between primary and secondary structure of proteins.
3. Name the Vitamins whose deficiency cause i) rickets ii) scurvy
4. Write the Zwitter ion structure of alanine

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